
High-Powered Computing (HPC) is rapidly becoming a cornerstone of technological advancements. HPC consume massive amounts of energy which leads to environmental consequences ranging from carbon emissions to water pollution.

In Part 1, we estimated the total energy consumed by HPC per year. We needed to make several assumptions to estimate this value, but all of them were well-founded and justifiable. After calculating, we found that HPC consumed about 324 TWh per year. This was an important figure that we referenced later on in our paper. We continued the analysis of energy consumption by HPC by considering the full capacity of global electricity production (which powers data centers in which HPC takes place) and the average utilization rate of HPC facilities. Two percentages were calculated: one was the percentage of current global electricity consumed by HPC, and the other was the percentage of global electricity consumed by HPC in a hypothetical reality where electricity production and HPC performance were perfectly efficient. Although, of course, this second reality is not possible, it allowed us to gauge the capability of global energy-producing facilities and how much growth in the HPC industry they allow for.

Our model to measure the environmental impact of HPC usage was a simplified version of the BernSCM, an established climate model that is used to determine the spread of carbon content after emissions into the atmosphere as well as temperature increase. We used the results of our altered version of BernSCM to also determine effects in terms of ocean acidification and sea level rise. In our development of our models, we utilized scientific reasoning to determine how carbon was transferred between the atmosphere, ocean, and land biosphere, as well as empirical results from established studies that report relationships between data such as a linear correlation between sea level rise rate and temperature change. We also utilized impulse response functions to help account for the delayed response in the biosphere in returning carbon that enters it through NPP. We coded our model in Matlab, with our code recreated in our paper. Our model is easily extendable if additional factors were to be considered, such as if it was to be used to predict emissions effects on larger timescales. We also developed a model the environmental impact of water usage of HPC that quantified how water consumption was directly linked to energy usage. The model took into various factors like recycling rate of water and efficiency of water use. Through this consumption model, we then created a pollution model, measuring the total mass of pollutants discharged. These two models allowed us to assess the environmental consequences of HPC operations.

When considering only HPC energy usage into 2030 we determined that atmosphere carbon content would increase by about 1.64 GtC, ocean carbon by .01 GtC, land biosphere carbon by 1.86 GtC, and temperature by .012 °C. When considering HPC usage alongside the average annual carbon emissions worldwide, by 2030 atmosphere carbon increases by about 225 GtC, ocean carbon by 2.3 GtC, and land biosphere carbon by 1.86 GtC. In addition, sea level was projected to increase by 16 mm and a cumulative Δp_h in the ocean of -.051. These are by no means negligible effects, and it is clear we must take action to reduce the impacts of HPC energy usage through increased efficiency and renewable energy use to minimize carbon emissions. We also developed a model to determine the environmental impact of water usage of HPC.

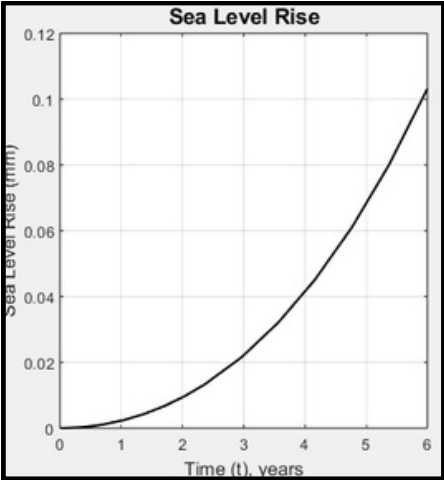
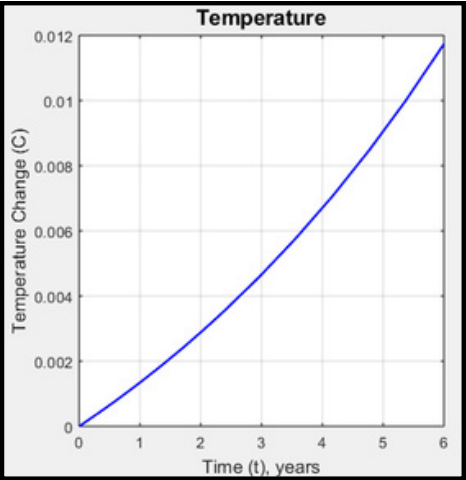
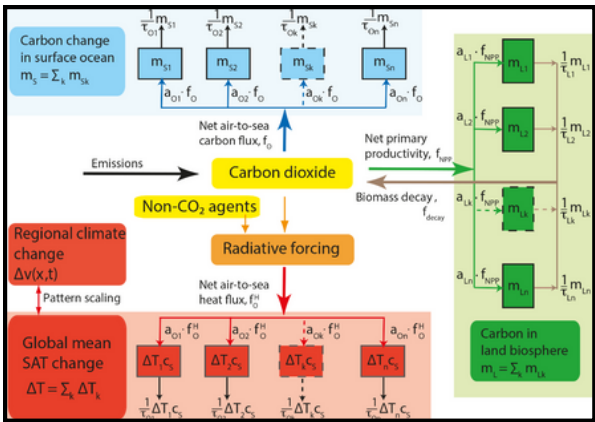
Our analysis highlights the need to adopt sustainable technology to minimize the environmental impacts of HPC facilities.

Dear United Nations Advisory Board,

Recently, high-powered computing (HPC) has taken a center stage in technological advancement, but we must also address the environmental concerns of it. The Advisory Board's recent report, "Governing AI for Humanity," focuses on the ethical considerations surrounding AI, however it does not sufficiently address the environmental impact of HPC, specifically in its carbon footprint and water usage. This oversight is concerning, and we urge that in any future reports regarding HPC that its environmental impacts be highlighted. As a collective, we must increase our awareness of this issue to avert any catastrophic consequences in the future.



Our team has conducted extensive research and is eager to share our findings regarding the environmental impact of HPC. We believe that the upcoming goals for 2030 provide a critical opportunity to address this challenge by incorporating strategies for reducing the environmental impact of HPC into the global AI governance framework.



Our model measures impact in terms of carbon content in the atmosphere, ocean, and land biosphere, as well as measuring ocean acidification and sea level rise. To 2030, we have determined that given the rate of increase of HPCs, that global temperature will rise by .01 degrees C and sea level will increase by .1 mm, with more detailed results in our paper. These results are not negligible, and we urge that the Advisory Board includes environmental concerns in any future reports regarding HPC and related issues.

By increasing energy efficiency and developing infrastructure to support increased renewable energy usage, we can help combat the environmental issues that HPC presents. Additionally, by implementing heat reuse systems that capture and reuse waste heat generated by equipment to heat nearby buildings, could help to offset some of their carbon footprint. With a more detailed section on the environmental consequences and measures to address the environment, of HPC in future reports, the United Nations Advisory Board can set a proactive example for sustainable AI development.



Thank you for your trust in us.

Best,
Team 16158

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1 Introduction

“Technology is a useful servant but a dangerous master.” —Christian Lous Lange, 1921

Artificial intelligence. Genome sequencing. Self-driving cars.^[1] These are some of the most exciting advancements today, attracting boundless innovation and progressing human achievements in all aspects. But they all share one characteristic: being driven by high-powered computing (HPC).

Evidently, the rise of HPC has opened many doors, exponentially decreasing the time required to perform certain types of data analysis and calculation. However, this immense power comes at a cost. HPC consumes enormous amounts of energy to make all this possible, and this has potentially disastrous effects on the environment. A singularly impressive development is contributing to an existential threat to our survival.

The purpose of this paper is to more closely examine these environmental effects to guide judgment on how best to use this extraordinary new technology while consuming energy responsibly.

1.1 Problem Restatement

We are tasked with creating models that can accurately model the impact of increasing HPC usage on the environment and discussing the findings and extendability of our model.

1. First, we must evaluate the scope of the issue of HPC usage, and we do this by determining a yearly energy consumption rate taking into account both full capacity and energy utilization.
2. Then, we create a model that will allow us to determine measurable environmental impacts associated with this energy consumption (specifically carbon emissions from this consumption) worldwide. Our model will account for the different sources of energy used in HPC usage.
3. We will then apply our model to provide insight into how HPC growth will affect the environment up to the year 2030. We will also discuss how HPC growth and different energy mixes affect our model.
4. We will use our model to simulate conditions where HPC energy consumption uses more renewable energy sources, in turn reducing carbon emissions. We will also look into expanding our model beyond only the environmental impacts of carbon emissions.
5. Finally, we will discuss our findings and develop a plan of action that will minimize the negative environmental impact in a letter to the United Nations Advisory Board. We will use our model to justify this plan of action in hopes that the Board will take this into consideration and focus more on the environmental impact of HPCs.

2 Describing the Scope (Part 1)

2.1 Variable Definitions

Table 1: Variable definitions (Part 1)

Variable	Definition
E_{DC}	Total energy consumption of data centers
$E_{DC_{HPC}}$	Total energy consumption of data centers for HPC
E_{HPC}	Total energy consumption of HPC
$U_{DC_{NHPC}}$	Energy consumption of data centers per unit of computing not done by HPC
$U_{DC_{HPC}}$	Energy consumption of data centers per unit of computing done by HPC
C_{DC}	Amount of computing done in data centers
$C_{DC_{HPC}}$	Amount of computing done in data centers by HPC
R_{DC}	Revenue from data centers
$R_{DC_{HPC}}$	Revenue from data centers from HPC
PUE	Ratio of energy consumed by a data center to energy used to run IT equipment

2.2 Assumptions

1. We assume that $E_{DC_{HPC}}$ closely approximates the total energy consumption of HPC.

Justification: HPC was first developed in the context of research, but in recent years, some of the largest companies in the world have started taking advantage of its calculation capabilities.^[2] Currently, the volume of HPC taking place in data centers by these companies likely greatly outweighs the volume of HPC taking place in individual university and research facilities. In addition, although cryptocurrency firms have recently begun to utilize HPC, this usage also occurs in data centers; cryptocurrency mining used to be a household activity, but this is now impractical, as it would take nine years of household electricity to mine a single Bitcoin,^[3] the standard cryptocurrency of choice for mining.^[4] Moreover, Bitcoin is so volatile that it sometimes costs more to mine Bitcoin than it is worth.^[3] Based on this cost-benefit analysis, it is unlikely that many households would choose to mine Bitcoin, and many other cryptocurrencies have the same characteristics, so it is reasonable to assume that virtually all mining takes place in data centers.

2. We assume that the proportion $R_{DC_{HPC}}/R_{DC}$ closely approximates $C_{DC_{HPC}}/C_{DC}$.

Justification: Data centers bring in revenue by running computations. So, there must be a general positive relationship between revenue and volume of computing ($R_{DC} \propto C_{DC}$), and this relationship should extend to revenue brought in by HPC and volume of computing done by HPC, too ($R_{DC_{HPC}} \propto C_{DC_{HPC}}$). It stands to reason, then, that $R_{DC_{HPC}}/R_{DC} \approx C_{DC_{HPC}}/C_{DC}$. We acknowledge that this approximation may not be completely accurate if the computing jobs that HPC is used for tend to have higher or lower returns—in these cases, $R_{DC_{HPC}}/R_{DC}$ would overestimate and underestimate $C_{DC_{HPC}}/C_{DC}$, respectively.

2.3 Energy Consumption

One of the largest concerns about HPC is that it requires a significant amount of energy to run. This energy demand is only increasing—the International Energy Agency predicts that energy consumption by data centers is growing at 20 to 40% per year, driven by growing demand for HPC.^[5] This has potentially disastrous implications that we discuss later in the paper. To fully understand these implications, however, it is important to first quantify the current global level of HPC energy consumption, or E_{HPC} .

Limited data availability makes it difficult to find a value for E_{HPC} , but it is possible to estimate it. We were able to find that data centers consume about 460 TWh per year,^[6] that HPC accounts for 23% of the data center market,^[7] and that HPC consumes eight times more energy than “general-purpose computing,” which we assumed is the standard type of computing used in data centers.^[8] (It is important to note that this ratio of 8:1 is not $\frac{E_{DC_{HPC}}}{E_{DC} - E_{DC_{HPC}}}$ but rather $\frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}}$, since 8:1 is the ratio of $\frac{E_{DC_{HPC}}}{E_{DC}}$ per unit of computing, but we need to account for the relative amount of computing done by HPC.) So, we have:

$$\begin{aligned} E_{DC} &= 460 \text{ TWh}, \\ \frac{R_{DC_{HPC}}}{R_{DC}} &= 0.23, \text{ and} \\ \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}} &= 8. \end{aligned}$$

Also, through assumption 1, we have:

$$E_{HPC} = E_{DC} \cdot \frac{E_{DC_{HPC}}}{E_{DC}} = E_{DC_{HPC}}.$$

So, we want to find $\frac{E_{DC_{HPC}}}{E_{DC}}$. We have:

$$\frac{E_{DC_{HPC}}}{E_{DC}} = \frac{C_{DC_{HPC}} \cdot U_{DC_{HPC}}}{C_{DC_{HPC}} \cdot U_{DC_{HPC}} + (C_{DC} - C_{DC_{HPC}}) \cdot U_{DC_{NHPC}}}.$$

Our goal is to rearrange this fraction in terms of $\frac{R_{DC_{HPC}}}{R_{DC}}$ and $\frac{E_{DC_{HPC}}}{E_{DC}}$. To do this, we divide both the numerator and

denominator by $C_{DC} \cdot U_{DC_{HPC}}$ and apply our second assumption:

$$\begin{aligned} \frac{E_{DC_{HPC}}}{E_{DC}} &= \frac{\frac{C_{DC_{HPC}}}{C_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}}}{\frac{C_{DC_{HPC}}}{C_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}} + \left(1 - \frac{C_{DC_{HPC}}}{C_{DC_{NHPC}}}\right) \cdot 1} \\ &= \frac{\frac{R_{DC_{HPC}}}{R_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}}}{\frac{R_{DC_{HPC}}}{R_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}} + \left(1 - \frac{R_{DC_{HPC}}}{R_{DC_{NHPC}}}\right) \cdot 1}. \end{aligned}$$

Now, we can use this value for $\frac{E_{DC_{HPC}}}{E_{DC}}$ in the original equation for E_{HPC} and plug in our values:

$$\begin{aligned} E_{HPC} &= E_{DC} \cdot \frac{\frac{R_{DC_{HPC}}}{R_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}}}{\frac{R_{DC_{HPC}}}{R_{DC}} \cdot \frac{U_{DC_{HPC}}}{U_{DC_{NHPC}}} + \left(1 - \frac{R_{DC_{HPC}}}{R_{DC_{NHPC}}}\right) \cdot 1} \\ &= 460 \text{ TWh} \cdot \frac{0.23 \cdot 8}{0.23 \cdot 8 + (1 - 0.23) \cdot 1} \\ &= 324 \text{ TWh}. \end{aligned}$$

So, we estimate that HPC consumes a total of 324 TWh per year.

2.4 Full Capacity and Average Utilization Rates

As explained previously, global energy demands are only rising, and HPC will contribute to this rise in two ways: it will continue to advance technologically (therefore requiring more energy to operate), and it will continue to grow in popularity. Perhaps the most concrete example of this has been the growth of HPC in cryptocurrency mining—since October 2023, when Bit Digital became the first cryptocurrency mining firm to invest in HPC, many other top firms have invested billions of dollars into HPC.^[9] It is clear that HPC will only continue to demand more energy.

Further nuance in understanding the energy consumption of HPC can be gained by examining the energy consumption of HPC through the concepts of average utilization rate and full capacity. We begin by calculating the percentage of global electricity production (the type of energy used by data centers) that HPC consumes. Then, we find the average utilization rate of data centers, which we can equate to the average utilization rate of HPC (since $E_{DC_{HPC}} \approx E_{HPC}$ from assumption 1). After that, we find the proportion of the current global energy production needed to power HPC and the proportion of the current global energy *capacity* needed to power HPC. In the face of ever-increasing energy demands, especially for HPC, it is helpful to know the absolute extent to which our current energy infrastructure can support these demands in a hypothetical reality in which energy production and HPC were completely efficient.

Currently, in our inefficient reality, HPC consumes E_{DC} , which we found to be 324 TWh. In addition, current global electricity production is 25,500 TWh.^[10] So, HPC consumes $\frac{324 \text{ TWh}}{25,500 \text{ TWh}} = 1.27\%$ of total energy production.

Now, we can compute this same percentage in a hypothetical reality where HPC and energy production is completely efficient. The average utilization rate of data centers is equivalent to

$$\begin{aligned} \frac{1}{PUE} &= \frac{1}{(\text{Energy consumed by data center})/(\text{Energy used to run IT equipment})} \\ &= \frac{\text{Energy used to run IT equipment}}{\text{Energy consumed by data center}}. \end{aligned}$$

According to the National Renewable Energy Laboratory, the average PUE for data centers is around 1.8,^[11] so the average utilization rate is $\frac{1}{1.8} = 0.56$. We can multiply this by E_{DC} , or 324 TWh, to find the amount of energy that is ideally required to run HPC:

$$324 \text{ TWh} \cdot 0.56 = 181 \text{ TWh}.$$

Now, we can calculate the global energy full capacity. In 2022, global electricity generation capacity was 8.5 TW. Over the course of a year, this translates to:

$$8.5 \text{ TW} \cdot \frac{8760 \text{ hours}}{\text{year}} = 75000 \text{ TWh}.$$

This means that, in a hypothetical reality where energy production and HPC are completely efficient, HPC would consume just $(181 \text{ TWh})/(75,000 \text{ TWh}) = 0.24\%$ of total energy production.

This percentage is much less than the current percentage of 1.27%, which suggests that as technology improves and, therefore, efficiency improves (both for HPC and electricity production), HPC will become less and less of a threat to global energy supply. However, this does not account for the potential for the growth of energy consumption by HPC to outpace the growth of global electricity production. In fact, this seems probable—the HPC market is expected to grow at about 7.3% annually over the next several years,^[12] while global electricity production has historically grown by about 2.5% annually.^[13]

3 Modeling the Impact of Carbon Emissions (Part 2)

3.1 Variable Definitions

Table 2: Variable definitions (Part 2) (Note: 1 GtC = 1 gigaton of carbon = 10^{12} kg C)

Variable	Definition	Unit
X	Total amount of energy consumed	Free
a, b, c, d, e	Proportions of energy used by major fuel sources	Unitless
m_A	Carbon content in atmosphere	GtC
m_S	Carbon content in surface ocean	GtC
m_L	Carbon content in land biosphere	GtC
ΔT	Change in global surface air temperature (SAT)	$^{\circ}\text{C}$
e	(in climate model) rate of carbon emissions to atmosphere	GtC/yr
f_O	Air to ocean carbon flux	GtC/yr
k_g	Gas exchange coefficient for CO_2 , air to ocean (≈ 0.134)	$\text{yr}^{-1} A_O^{-1}$
A_O	Surface area of ocean ($\approx 3.5 \times 10^{14}$)	m^2
c_s	Heat capacity of ocean surface ($\approx 7.26 \times 10^{22}$)	J/K
ε	Atmospheric mass ratio of Carbon to mixed water CO_2 (≈ 2.123)	GtC/ppm
$p_A^{\text{CO}_2}$	Atmospheric CO_2 pressure	ppm
$p_S^{\text{CO}_2}$	Surface Ocean CO_2 pressure	ppm
f_{CO_2}	Ocean CO_2 Fugacity	ppm
T	Ocean Average Surface Temperature ($\approx 18.3^{\circ}$)	$^{\circ}\text{C}$
f_O^H	Air to ocean heat flux	W
RF, RF_{CO_2}	Radiative Forcing, change in energy balance in the atmosphere due to gas (used for GHGs)	W/m^2
ΔDIC	Change in inorganic dissolved carbon in ocean layer	$\mu\text{mol}/\text{kg}$
a_O	Ocean fraction of Earth Surface (≈ 0.71)	Unitless
ΔT^{eq}	Equilibrium ΔT for a given radiative forcing (GHG content)	$^{\circ}\text{C}$
$RF_{2x}, \Delta T_{2x}$	Equilibrium RF and ΔT at double the initial CO_2 content	$\text{W}/\text{m}^2, ^{\circ}\text{C}$
λ	Climate Sensitivity	$\text{W}/(\text{m}^2^{\circ}\text{C})$
$NPP(t)$	Net Primary Productivity in land biosphere	GtC/yr
$f_{\text{decay}}(t)$	Carbon released from biomass decay (death, fires, respiration)	GtC/yr
$r(t)$	Impulse Response Function (IRF)	N/A
$\delta(t)$	Dirac Impulse Generalized Function	N/A
H	Global Mean Sea Level	mm

3.2 Assumptions

1. **We assume that the percentage breakup of energy sources used for HPC energy consumption is approximately the same as that of the world.**

Justification: We are modeling HPC usage on a global scale, so approximating the proportion of each major energy type (coal, oil, gas, biomass, renewables) produced is useful in simplifying our model.

2. **We assume that renewable energy sources produce a negligible amount of CO_2 emissions.**

Justification: Many steps in the production of energy for renewable energy sources (including solar, wind, and hydro) use some form of nonrenewable energy, producing carbon emissions in the process. However, the vast majority of carbon emissions are from nonrenewable energy sources.

3. **We assume that the effects of deep sea carbon and heat transfer are negligible for the purposes of our model.**

Justification: Transfer of carbon from the surface ocean to the deep sea and heat transfer from the deep ocean to the atmosphere only occurs in non-negligible quantities on the timescale of decades to millennia. Since we are attempting to predict up to 2030, these effects are negligible in the context of our model.

4. **We assume that the effects of continental heat transfer are negligible for the purposes of our model.**

Justification: Similarly, the effects of continental heat transfer on the global surface temperature ΔT are negligible in the context of our model.

5. **We assume that we can treat the atmosphere, ocean surface, and land biosphere as relatively spatially homogeneous, not taking into account climate intricacies around the world.**

Justification: Modeling climate intricacies around the world would be too complicated. Assuming homogeneity allows us to treat each sector as a whole in our model. In general, our results indicate worldwide effects of carbon emissions due to HPCs, but it may not be as accurate for particular areas where conditions are abnormal.

6. **We assume that efficiency of conversion of each energy source (including coal, oil, and gas) to energy is constant**

Justification: Though we do not optimize extracting energy from fossil fuels to minimize carbon emissions, we assume that over the short time period we are considering (up to 2030), rates of conversion do not change significantly. This also helps us to simplify our model in determining HPC carbon emissions.

3.3 Energy Consumption to Carbon Emissions

We are tasked with creating a model to determine the environmental impacts of the energy consumption of HPC. We have already determined the annual energy consumption associated with HPC, so we first find how much carbon dioxide is emitted into the atmosphere given some amount of energy consumption.

We assume that the breakup of energy production for HPC is analogous to the worldwide breakup for all energy demand. This changes significantly depending on the specific location, but since we are considering HPC worldwide, we make this assumption. We consider the major energy sources that contribute to carbon emissions, namely coal, gas, oil, and biomass. All energy produced is either one of these four types or from some method that we say has negligible impact on carbon emissions. In 2023, around 82.6% of energy produced worldwide was from one of these major sources. Around 6% was from biomass, 24.9% from coal, 29.8% from oil, and 21.9% from natural gas.

First, our model takes an input to represent the current energy mix, with the proportion of coal, oil, gas, biomass, and other sources being a, b, c, d , and e , respectively, such that

$$a + b + c + d + e = 1.$$

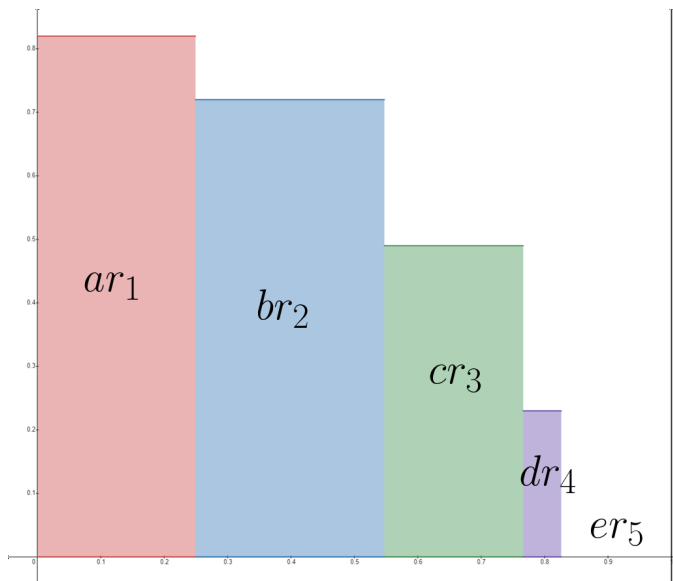
Then, we multiply by the total amount of energy consumed, X , to get

$$X + bX + cX + dX + eX = X.$$

Now, each term on the left side represents the amount of energy created by each source. Then, for each term, we multiply by the amount of CO_2 released per unit energy (like TWh) and add those all up to get the total carbon emission rate. Let each of those rates be r_i for $1 \leq i \leq 5$, but we always take $r_5 = 0$ to represent energy sources with negligible carbon emissions. This can be visualized on a graph where the amount of CO_2 emitted per unit energy is on the y -axis and the proportion of energy produced by each source on the x -axis. Once we have found the amount of carbon emissions (over some time period), we then model the environmental impact of this carbon.

3.4 Altered BernSCM

To determine the impacts of the CO_2 deposited into the atmosphere, we decided to use a simplified version of the Bern Simple Climate Model (BernSCM)^[15] to model the carbon transfer between the atmosphere, ocean, and land biosphere. Carbon that enters the atmosphere from emissions will dissipate to the land biosphere and ocean. Then, we can determine the change in the global surface air temperature (Global SAT) by modeling the heat transfer from the surface ocean and land biosphere. Increased atmospheric CO_2 increases photosynthesis in plants, so we model the carbon transfer to the land biosphere using net primary productivity (NPP). Similarly, the land biosphere returns carbon to the atmosphere



(a) Proportions of Energy produced, each area is kg of CO₂ per kWh consumed by HPCs

Energy Type	kg CO ₂ emitted per kWh
Coal	.82 kg
Oil	.72 kg
Natural Gas	.49 kg
Biomass	.23 kg

(b) kg of CO₂ per kWh per energy source^[14], y -axis of graph to the left

Figure 1: Visual Representation of converting Energy Usage to Carbon Emissions

through respiration, biomass decay, fires, and other natural processes. BernSCM also takes into account deep sea carbon and heat transfer, but we will not implement this in our model.

BernSCM splits each sector (temperature, ocean, land biosphere) into multiple different boxes that, in practice, can be used to represent different parts of the land biosphere/ocean that function differently with respect to carbon or heat transfer. The main quantities of interest are m_A , m_S , m_L , which are carbon stored in the atmosphere, ocean, and land biosphere respectively, and ΔT , the global SAT change.

3.5 Model Setup

We will model the transfer between the atmosphere, land biosphere, and ocean with differential equations. For m_A , we have

$$\frac{dm_A}{dt} = e - \frac{dm_S}{dt} - \frac{dm_L}{dt}$$

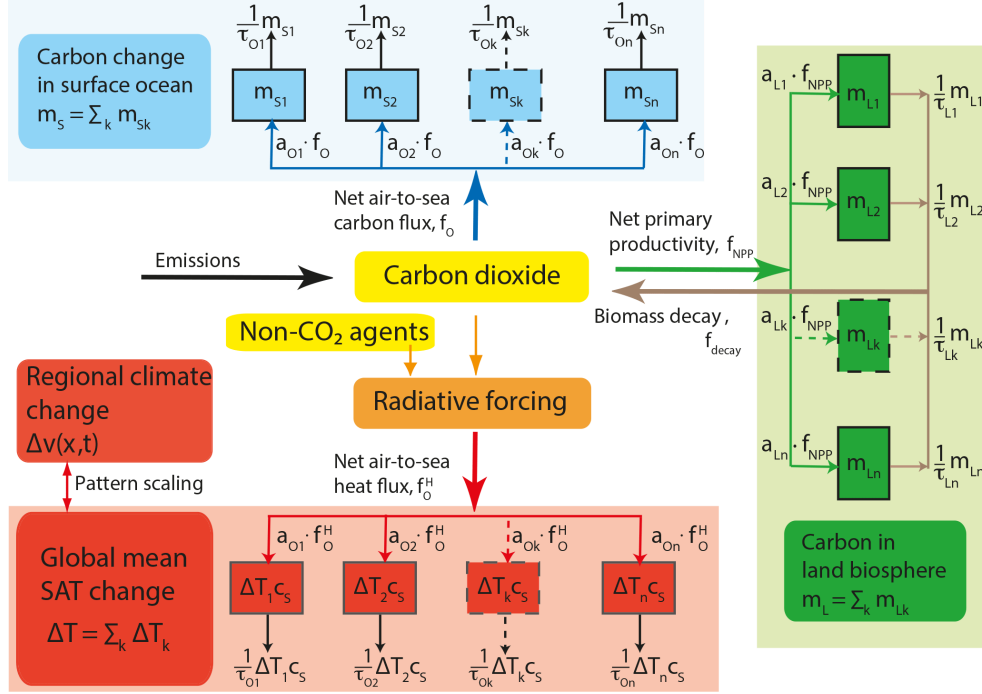
Here, e is the rate of carbon emissions. This equation means the rate of change of atmospheric carbon is equal to the rate of emissions entering minus the rate of change of land biosphere and ocean carbon, since the atmosphere transfers carbon to both the ocean and land biosphere. Human activities are not considered in the land biosphere, which is why we have e to represent emissions.

3.5.1 Surface Ocean Carbon

Since we are ignoring deep sea transfer, we have that the change in carbon levels in the ocean is the flux of carbon from the atmosphere to the surface ocean, or $\frac{dm_S}{dt} = f_O$. This flux is proportional to the CO₂ pressure difference between the air and the surface ocean. We can find that the constant of proportionality takes into account to the rate of CO₂ exchange between atmosphere and ocean, worldwide surface area of the ocean, and converting between CO₂ to carbon mass. We take these constants directly from what was used in implementation of the BernSCM.

$$\frac{dm_S}{dt} = f_O = k_g A_O \varepsilon (p_A^{CO_2} - p_S^{CO_2}) \approx 1.007 \times 10^{14} (p_A^{CO_2} - p_S^{CO_2}).$$

For ocean partial pressure of CO₂, we first chose to use a fitted equation for fugacity (partial pressure adjusted for non-ideal gas conditions) of CO₂.^[17] This provides an equation for $\ln f_{CO_2}$ that is quadratic with respect to temperature and has relatively low error compared to measured data. However, in running our model we ran into an issue since the magnitude of fugacity by these calculations was too low to control ocean carbon growth. So, we instead opted to adopt BernSCM's

Figure 2: Graphical display of BernSCM.^[15]

calculation of $p_S^{CO_2}$ using dissolved inorganic carbon change in the ocean (ΔDIC). To find $p_S^{CO_2}$, we have

$$\begin{aligned} \Delta DIC &= 4.587 m_s \\ p_S^{CO_2} &= 1.3 \Delta DIC + 3.77 \times 10^{-3} \Delta DIC^2 + 0.924 \times 10^{-5} \Delta DIC^3 + \\ &\quad 0.136 \times 10^{-7} \Delta DIC^4 + 1.258 \times 10^{-10} \Delta DIC^5 \end{aligned}$$

This equation was found from fitting data from global average temperature and ΔDIC to $p_S^{CO_2}$. We also have that $p_A^{CO_2} = m_A \epsilon^{-1}$, since the atmospheric pressure times the mass ratio of carbon to CO_2 is just the carbon content. So, we have that the change in ocean carbon is a function of atmosphere content and change in temperature, which can be expressed as

$$\frac{dm_S}{dt} = 7.51 \times 10^{14} m_A - 1.007 \times 10^{14} p_S^{CO_2}$$

3.5.2 Change in Global Surface Air Temperature

We will also model the change in global SAT to determine a general measure for climate change as a result of carbon emissions. This will help us determine some central impacts of the emissions due to HPC. As mentioned in previous sections, we ignore deep sea heat uptake, and continental heat uptake is negligible relative to ocean heat update. [1] also provides a formula to represent this. We have

$$\frac{d\Delta T}{dt} c_s = f_o^H = RF \left(1 - \frac{\Delta T}{\Delta T^{eq}} \right) \frac{A_O}{a_O}$$

Here, RF is the radiation forcing factor or change in radiative flux. Addition of carbon to the atmosphere disrupts radiative/thermal equilibrium, which then changes equilibrium temperature (T^{eq}) due to an increased carbon content in the atmosphere and ocean. The radiation forcing factor changes based on the different gases present in the atmosphere. Equilibrium temperature given a certain RF will allow us to find the change in temperature. The factor of $\frac{A_O}{a_O}$ is the surface area of the Earth, as A_O is the surface area of the ocean and a_O is the proportion of the Earth's surface covered by ocean. Scaling by this allows us to find an average overall temperature change worldwide.

To approximate the ΔT^{eq} and RF , we utilize the concept of climate sensitivity. The climate sensitivity is a measure of the temperature increase per doubling of CO_2 , and it is a constant that can be expressed as $\lambda = \frac{RF_{2x}}{\Delta T_{2x}}$. RF_{2x} and ΔT_{2x} is the RF and ΔT when atmospheric CO_2 is at twice the initial conditions (m_{A0}). A doubling standard is used to represent a significant change from the present and is a good baseline to use to advise change with respect to it. The climate

sensitivity is a metric for how the atmosphere responds to an increase in carbon (or any other gases), and λ is a constant that allows us to find ΔT^{eq} given some RF . RF can be approximated to be the RF due to CO_2 concentration (as we have assumed no change in other substantial GHG concentrations or other influential conditions). We have

$$\Delta T^{eq} = \frac{RF_{CO_2}}{\lambda}$$

$$RF_{CO_2} = \ln \left(\frac{m_A}{m_{A0}} \right) \frac{RF_{2x}}{\ln(2)}$$

Plugging these into our equation for $\frac{d\Delta T}{dt}$, we get

$$\frac{d\Delta T}{dt} c_s = \frac{A_O}{a_O} \left(\ln \left(\frac{m_A}{m_{A0}} \right) \frac{RF_{2x}}{\ln(2)} - \lambda \Delta T \right)$$

Plugging in constant values for c_s , A_O , a_O , RF_{2x} , and λ (taken from the BernSCM implementation) this becomes

$$\frac{d\Delta T}{dt} \approx 3.67 \times 10^{-8} \ln \left(\frac{m_A}{m_{A0}} \right) - 8.44 \times 10^{-9} \Delta T$$

This only accounts for increase in temperature due to ocean carbon content, but there is also an effect due to atmospheric concentration as CO_2 in the atmosphere causing greenhouse effects, depleting the ozone layer and causing temperature increase. Research^[16] states that on a short time scale a linear approximation between carbon increase in the atmosphere and temperature, with a rate of about 1 degree celsius per 140 GtC added to the atmosphere. We add this to our model as

$$\frac{d\Delta T}{dt} = 3.67 \times 10^{-8} \ln \left(\frac{m_A}{m_{A0}} \right) - 8.44 \times 10^{-9} \Delta T + \frac{1}{140} \frac{dm_A}{dt}$$

3.5.3 Land Biosphere Carbon

For land biosphere concentration, we have

$$\frac{dm_L}{dt} = NPP(t) - f_{decay}(t)$$

Where $NPP(t)$ is the NPP and $f_{decay}(t)$ is the carbon released from biomass decay. Note that these are both functions of time. We will use Keeling's Formula,^[18] which describes a logarithmic dependence of NPP on CO_2 and has been verified by experimental data. We have

$$NPP(t) = NPP(0) \left(1 + \gamma \ln \left(\frac{m_A}{m_{A0}} \right) \right)$$

The proportion value $\gamma = 0.2$ has been found to fit global climate models.^[18] $NPP(0)$ and m_{A0} are initial values.

It is harder to find an expression for f_{decay} , since this directly relates to $NPP(t)$ which is changing over time due to carbon emissions. The effect of the change of NPP is delayed until a plant dies, so we can't have a direct dependence of f_{decay} on the current state of carbon in the land biosphere. To deal with this concern, we will use an impulse-response function (IRF).

3.5.4 Impulse Response Functions

An impulse response model of some function $g(t)$ can be expressed as

$$g(t) = \int_{-\infty}^t f(t') r(t - t') dt'$$

Here, $f(t)$ is the function representing addition to the system, and $r(t)$ is the impulse response function (IRF) that implements the response to the change in $f(t)$. For $g(t) = m_L(t)$, we will use $f(t) = NPP(t)$ since this is the addition of carbon to the system. $f(t')$ provides the impulse at time t' , and when we evaluate at t we are looking at the cumulative effect of all impulses before time t . Using an impulse response model to model biomass decay assumes that the system is time invariant and linear, using the time-variant function $f(t)$ to determine the change in a quantity some amount of time. In reality, decay like this is not linear since there are compounding effects across the atmosphere and ocean, which would be an improvement of this model.

In this case it is useful to think of $f(t')$, for some fixed input t' , as a Dirac impulse $\delta(t')$. The Dirac generalized function represents an instantaneous impulse and is used to solve for $r(t)$ by modeling the system with a homogeneous

differential equation. BernSCM implements impulse response functions and solves for the IRF by discretizing the land biosphere into a four box model, each representing a different delayed response rate. Generally, IRFs are solved for using a linear differential equation, which results in an exponential decay equation for $t \geq t'$. We will use the constants from BernSCM's implementation on four boxes, getting the function

$$r(t) = -1.57e^{-\frac{t}{2.18}} + 2.01e^{-\frac{t}{2.86}} + 0.268e^{-\frac{t}{20}} + 0.293e^{-\frac{t}{100}}$$

For the land biosphere, there ends up being no 'residual' carbon left after an arbitrarily long amount of time, since it will all end up flowing out of the land biosphere through natural means in a shorter amount of time. In each of the $e^{-\frac{t}{k}}$ factors, k represent the timescale of the decay in years.

We find the rate of change of m_L by taking the time derivative of the response integral equation above, getting

$$\begin{aligned}\frac{dg(t)}{dt} &= f(t) + \int_{-\infty}^t f(t') \frac{dr(t-t')}{dt} dt' \\ \frac{dm_L}{dt} &= NPP(t) + \int_{-\infty}^t NPP(t') \frac{dr(t-t')}{dt} dt' = NPP(t) - f_{decay}(t)\end{aligned}$$

Since we have a function for $NPP(t)$ and $r(t)$, we are able to numerically solve for $\frac{dm_L}{dt}$. This closes our system, since now we have found forms for each change in carbon content that depend only on initial inputs and time.

3.6 Effect of Carbon Emissions on the Environment

As carbon emissions continue to affect our environment, we are tasked with understanding the grave implications of high levels of carbon emissions. To understand the vast impact of carbon emissions, we will be looking at the environmental impact that we can directly and indirectly observe from BernSCM.

Our model gives us two major results: change in carbon content between the atmosphere, land biosphere, and ocean as well as the change in the global surface air temperature. By modeling these changes, we gain a powerful tool for understanding environmental impacts. This approach allows us to track carbon fluxes and quantify the effects of HPCs' carbon emissions. It also sheds light on how natural processes, like photosynthesis and oceanic carbon absorption, contribute to balancing these emissions. Simultaneously, tracking surface air temperature provides a direct link between these carbon changes and their impact on global warming.

In addition, we can utilize the results from this model to further understand major effects caused directly from CO_2 including rising sea levels and ocean acidification.

3.6.1 Rising Sea Levels

Rising sea levels occur as a result of increasing global temperatures, primarily driven by climate change. This phenomenon is significant as it threatens coastal communities, amplifies flooding, and erodes habitats for both humans and wildlife. While the main contributors are melting ice sheets and the thermal expansion of seawater as it warms, we will simplify the model by considering a general equation that represents rising sea levels directly as a function of change in temperature. This approach allows us to focus on the relationship between temperature and sea-level elevation without isolating specific contributing mechanisms.^[23]

In Rahmstorf's research, he discusses that rising sea levels must be reduced to a more simplistic model that assumes proportionality between the SAT and sea-level rise. This is modeled with the following:

$$\frac{dH}{dt} = a(T - T_0)$$

From this, we can write:

$$\frac{dH}{dt} = a\Delta T$$

Rahmstorf also finds the proportionality constant, a , with value 3.4mm/year per Celsius. We will use the same proportionality constant in our models as we assume that over a relatively short period of time, the system stays proportional. We can now write the following to find an adjusted height (in mm):

$$\frac{dH}{dt} = 3.4\Delta T$$

3.6.2 Ocean Acidification

Ocean acidification is a phenomenon where the pH levels of ocean waters decrease, becoming more acidic due to the absorption of excess carbon dioxide (CO_2) from the atmosphere. This process is significant because it disrupts marine ecosystems, harming calcifying organisms like corals, mollusks, and plankton that rely on carbonate ions to form shells and skeletons. As CO_2 dissolves in seawater, it reacts to form carbonic acid, which lowers pH and reduces the availability of these essential ions. We will understand the importance of carbon flux in the representation of ocean acidification.

We will be applying the assumption that the global change in marine carbon dioxide is directly correlated to the change in pH levels of the global ocean. This assumes a uniform system and that external factors will not impact this system. This extremely simplifies our model, allowing us to easily extrapolate the pH level of the ocean based on the carbon content.

First, we want to be able to quantify a relationship between time and the carbon content in the ocean in order to relate carbon content to pH.

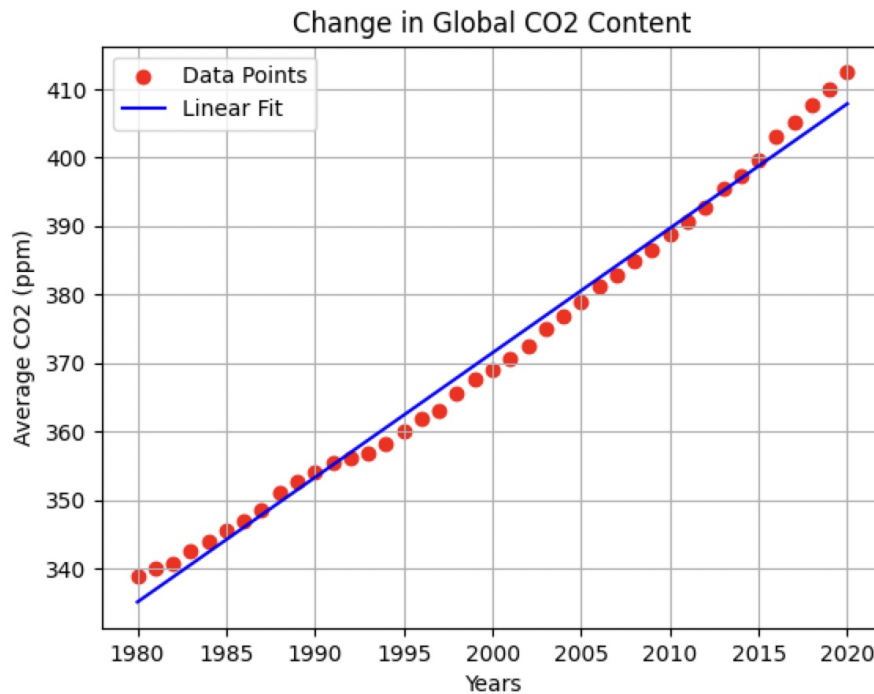


Figure 3: Displays yearly change in CO_2 content. Line described by $y = 1.82x - 3263.98$

Based on data comparing the average CO_2 levels in the oceans globally to time, we can observe that a linear regression fits the data points well with an R^2 value of 0.99. Since we can represent the global CO_2 levels as a line, this means that the rate of change of the global CO_2 content is constant and that any difference between constant time intervals are equivalent.

Now, we will consider the changes in pH levels of the ocean worldwide. From 1995 to 2020, the pH level dropped from 8.1 to 8.05^[25]. Using this data, we can correlate it to the change in CO_2 . Since this is a 25 year time frame, we find that the CO_2 content would have changed by 45.5 ppm. This means that for a -0.05 change in pH, the CO_2 content will have increased by 45.5 ppm. If we let x be the change in carbon content (in ppm), we can set up the following equation:

$$\Delta \text{pH} = -0.0011x$$

4 Applying Our Model (Part 3)

First, we calculate the total carbon emissions from HPC. From our calculations in Part 1, HPC facilities consume about 324 TWh of energy per year, and this is likely to increase over the following years. First, we will use the global distribution of energy production sources discussed in Section 3.3 to apply our climate model to, then we will see what happens for different energy distributions.

Our model will be working in GtC, or gigatons of carbon, so we will find the number of gigatons of carbon emissions from each energy source for 324 TWh. For coal, we have

$$\text{Emissions from Coal} = (0.249)(0.82 \frac{\text{kg}}{\text{kWh}})(324 \text{ TWh}) \left(\frac{10^9 \text{ kWh}}{\text{TWh}} \right) \left(\frac{10^{-12} \text{ GtC}}{\text{kg C}} \right) = .0662 \text{ GtC}$$

This is equivalent to ar_1X from the discussion in Section 3.3, adjusted for units. We perform the same calculations for other sources:

$$\text{Emissions from Oil} = (0.298)(0.72 \frac{\text{kg}}{\text{kWh}})(324 \text{ TWh}) \left(\frac{10^9 \text{ kWh}}{\text{TWh}} \right) \left(\frac{10^{-12} \text{ GtC}}{\text{kg C}} \right) = .0695 \text{ GtC}$$

$$\text{Emissions from Natural Gas} = (0.219)(0.49 \frac{\text{kg}}{\text{kWh}})(324 \text{ TWh}) \left(\frac{10^9 \text{ kWh}}{\text{TWh}} \right) \left(\frac{10^{-12} \text{ GtC}}{\text{kg C}} \right) = .0348 \text{ GtC}$$

$$\text{Emissions from Biomass} = (0.06)(0.23 \frac{\text{kg}}{\text{kWh}})(324 \text{ TWh}) \left(\frac{10^9 \text{ kWh}}{\text{TWh}} \right) \left(\frac{10^{-12} \text{ GtC}}{\text{kg C}} \right) = .00447 \text{ GtC}$$

Summing these up, we get that HPCs produce 0.1749 Gt of Carbon in a year they consume 324 TWh of energy. The rate we will input into our climate model for e will be global carbon emissions plus this, and we will account for HPC growth as exponential over time.

We coded the climate model in Matlab to numerically solve for the system of differential equations. We ran into trouble fully implementing numerical solutions using the impulse response function at each time step, but our results still apply to the system. The differential equation system we implemented is as follows, with explanations for equations from the previous section:

$$\begin{aligned} \frac{dm_A}{dt} &= e - \frac{dm_S}{dt} - \frac{dm_L}{dt} \\ \frac{dm_S}{dt} &= 7.51 \times 10^{14} m_A - 1.007 \times 10^{14} p_S^{CO_2} \\ \frac{d\Delta T}{dt} &= 3.67 \times 10^{-8} \ln \left(\frac{m_A}{m_{A0}} \right) - 8.44 \times 10^{-9} \Delta T + \frac{1}{140} \frac{dm_A}{dt} \\ \frac{dm_L}{dt} &= NPP(t) + \int_{-\infty}^t NPP(t') \frac{dr(t-t')}{dt} dt' \end{aligned}$$

Where we define

$$\begin{aligned} p_S^{CO_2} &= 1.3\Delta DIC + 3.77 \times 10^{-3} \Delta DIC^2 + 0.924 \times 10^{-5} \Delta DIC^3 + \\ &\quad 0.136 \times 10^{-7} \Delta DIC^4 + 1.258 \times 10^{-10} \Delta DIC^5 \\ \Delta DIC &= 4.587 m_s \\ NPP(t) &= NPP(0) \left(1 + 0.2 \ln \left(\frac{m_A}{m_{A0}} \right) \right) \\ r(t) &= -1.57e^{-\frac{t}{2.18}} + 2.01e^{-\frac{t}{2.86}} + 0.268e^{-\frac{t}{20}} + 0.293e^{-\frac{t}{100}} \end{aligned}$$

We found our initial values m_{A0} , m_{S0} , m_{L0} and $NPP(0)$ through research, getting $m_{A0} = 3310 \text{ GtC}$,^[20] $m_{S0} = 145.8 \text{ GtC}$,^[21] $m_{L0} = 470 \text{ GtC}$,^[19] and $NPP(0) = 62.1 \text{ GtC}$.^[22]

To model the growth of HPC, we decided to make the rate of emissions increase exponentially by a factor of 1.15 (or 15%) per year, as the HPC market is only going to keep growing now. Modeling the growth as exponential allows us to see the direct environmental impacts if we do not try to control emissions from HPCs. First, we ran our model with only emissions from HPC energy consumption to see the general effects. Then, we ran it alongside the global carbon emissions from all sources to see the effect HPC growth had. Both simulations were run for the next 6 years, which takes us into 2030.

Interestingly enough, over time carbon contents and temperature seem to be increasing linearly. We can see that, even in a short time span, overall carbon emissions will cause a sizable increase in temperature of about 1.6 °C. The carbon concentration in the atmosphere, ocean, and land biosphere seem to increase linearly over time, but it is clear that carbon emissions majorly affect atmospheric carbon, then ocean and land biosphere carbon. The rate of growth of each of these with respect to time appears linear since we are only exponentially increasing HPC emissions (which make up a very small amount of total emissions) by a small factor of 15%. In reality, total carbon emissions are changing in complicated

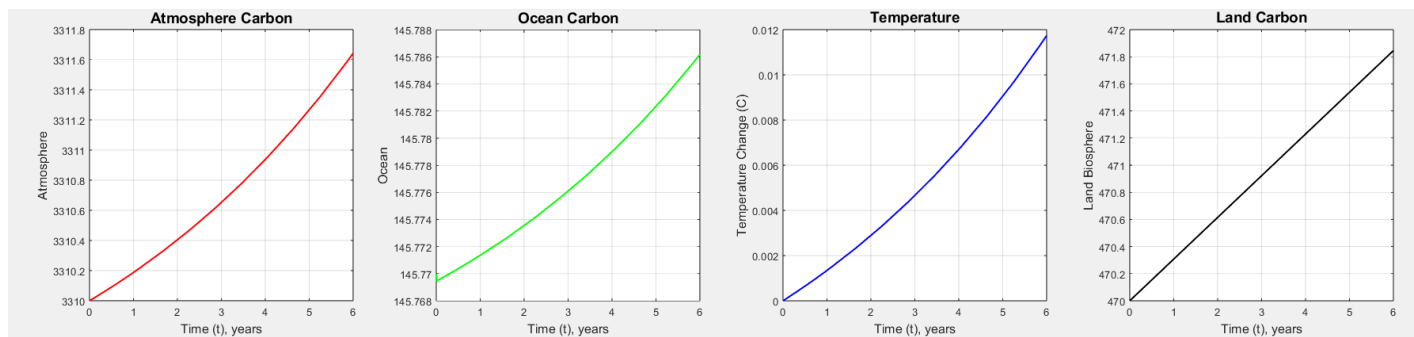


Figure 4: Carbon content/Temperature increase based only on HPCs

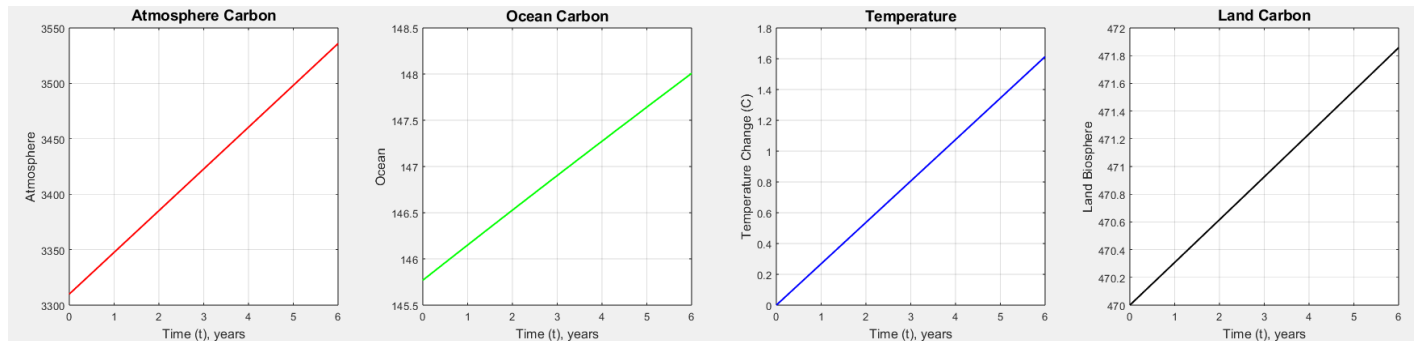


Figure 5: Carbon content/Temperature increase based on all Carbon emissions

ways, so depending on that change with respect to time we would see different results in the graph. Observing the GtC increase for just HPCs, into 2030 atmosphere carbon content increases by about 1.64 GtC, ocean carbon by .01 GtC, land biosphere carbon by 1.86 GtC, and temperature by .012 °C. When considering all carbon emissions and HPC, by 2030 atmosphere carbon increases by about 225 GtC, ocean carbon by 2.3 GtC, and land biosphere carbon by 1.86 GtC. This also shows that there is a relatively constant rate of increase to land biosphere independent from the carbon contents in the air and ocean, which could be justified by the rate that plants take in carbon.

To understand the further lingering effects of the carbon flux, we are able to apply results from our model to analyze ocean acidification. Applying 3.6.2, we know that since the carbon content in the ocean changes by 2.3 GtC, the Δ pH is -0.00164. This is a 13% increase from the -0.012 change in pH that we expect every six years from current carbon emissions. When we consider the full impact of carbon flux, this value increases greatly. From research, we know that approximately 31% of atmospheric CO₂ is absorbed into the ocean sink. Taking this into account, the full impact of the current carbon emissions come out to around 72 GtC. When the ocean sink has fully absorbed the CO₂ emitted in those six years from HPCs, we would expect a pH change of around -0.051. This value is much more significant, although of course, this neglects the time it will take for the ocean sink to fully absorb the carbon emissions.

We also implemented sea level rise as a measurable impact into our model governed by the equations in Section 3.6.1. As sea level growth increases linearly with change in temperature, we observe an exponential growth of sea level (since temperature increases linearly in our model). Applying our results, we got that HPC usage from now to 2030 will result in a 0.1 mm increase in sea level worldwide, and incorporating all carbon emissions we get a 16 mm increase. This increase could be due to associated factors like melting glaciers and thermal expansion of the ocean, as the relationship we found in research was found through regression. Sea level rise can lead to many detrimental effects such as saltwater intrusion and shoreline erosion, impacting our lives and biodiversity around the world. This is another central effect of carbon emissions and global warming; Though 16mm seems like a small increase, sea level height seems to be increasing exponentially, and we must step in and attempt to regulate this change.

4.1 Sensitivity Analysis

We are also tasked with exploring how our model reacts to different energy mixes and rates of growth of HPC and its energy consumption (or, testing sensitivity analysis on our model). For the sake of exploration, we will explore a few cases with different energy mixes and different HPC growth rates. Results are summarized in the table. For each of these, we will use initial carbon emissions rate per year of HPC to be 324 TWh. Since we have found that land biosphere carbon intake seems to depend only on time, we will omit those results from our sensitivity analysis (as they are all to 2030).

Table 3: Sensitivity Analysis Results

Energy Breakdown (%)					HPC Growth	Results (Increase due to HPC by 2030)		
Coal	Oil	Gas	Biomass	Other	Growth Rate	Atmosphere	Ocean	Temperature
0.80	0.20	0	0	0	12%	2.23 GtC	.023 GtC	.016 °C
0.80	0.20	0	0	0	50%	6.67 GtC	.07 GtC	.05 °C
0.4	0.1	0.3	0.1	0.1	15%	1.74 GtC	.01 GtC	.0124 °C
0.2	0.09	0	0.01	0.5	15%	.72 GtC	.01 GtC	.005 °C
0.2	0.09	0	0.01	0.5	50%	1.92 GtC	.02 GtC	.014 °C
0.1	0	0.1	0.3	0.5	20%	.71 GtC	.007 GtC	.005 °C
0.3	0.1	0.1	0.2	0.3	20%	1.51 GtC	.015 GtC	.01 °C

More analysis would include observing effects given different types of HPC growth (linear growth, asymptotic growth, etc). In general, we can see the effect that is made from switching from emission intensive fuel sources like coal and oil to renewables or even biomass. And, though it is fairly obvious, increasing the growth rate of HPC energy consumption over this time period will increase carbon content in the atmosphere and ocean and temperature increase. Some ways to improve this sensitivity analysis would be to take more factors into account like types of renewable energy (and their real carbon footprints), changes in efficiency of fuel source to energy output, and attempting to model longer into the future. Our model was developed to predict up to 2030, and we have made decisions that make it less viable for long time frames (for example, removing the deep sea as a carbon sink). So, we don't believe this model would fare well predicting over longer time periods, and long term carbon storage is a possible addition that would account for this.

5 Expansion of Model (Part 4)

5.1 Increasing Renewable Energy as a Source

We'd like to investigate the impact of increasing HPC usage of renewable energy sources for energy consumption on carbon emissions. We will do this by considering the initial distribution of energy sources, and as we increase renewable energy sources (in 'Other') we will decrease each of the main sources proportionally. The results are shown in the table below.

Table 4: Renewable Energy Increase

Energy Breakdown (%)					HPC Growth	Results (Increase due to HPC by 2030)		
Coal	Oil	Gas	Biomass	Other	Growth Rate	Atmosphere	Ocean	Temperature
0.249	0.298	0.219	0.06	0.174	15%	1.64 GtC	.016 GtC	.012 °C
0.241	0.289	0.212	0.058	0.2	15%	1.59 GtC	.016 GtC	.011 °C
0.181	0.216	0.159	0.044	0.4	15%	1.2 GtC	.012 GtC	.0085 °C
0.151	0.180	0.133	0.036	0.5	15%	1 GtC	.01 GtC	.0071 °C
0.075	0.09	0.066	0.018	0.75	15%	.47 GtC	.004 GtC	.0033 °C
0	0	0	0	1	15%	0 GtC	0 GtC	0 °C

Of course, carbon emissions aren't actually zero when switching to 100% renewable energy. Our model treats renewable energy sources as having zero carbon emissions associated with them, while there is still some (that depend on the type of renewable energy source). However, these results give us a good idea of the benefits of switching to full renewable energy, and that it will greatly reduce the environmental impact of HPC. How realistic is this? The main problem with implementing high levels of renewable energy usage is that they are not established enough to be used reliably like coal and oil are. Some, like solar or wind, rely on some weather condition to produce energy, so in emergencies it is not certain these are reliable. In addition, we are lacking infrastructure to support renewable energy sources, as HPC would require power grid support to take renewable energy worldwide. Some forms of renewable energy are not viable in different places in the world, unlike coal or oil. Though increasing renewable energy usage to 100% is an appealing choice to reduce carbon emissions, in the present it is not very feasible, and we should look towards hybrid options.

5.2 Variable and Constants Definitions

Table 5: Variable definitions

Variable	Definition
$W_{total}(t)$	Cumulative water consumed by HPC facilities up to time t
$W_{site}(t)$	Cumulative water consumed on-site by HPC facilities up to time t
$W_{source}(t)$	Cumulative water consumed by electricity utility used to generate the energy supplied to the HPC up to time t
$E_{total}(t)$	Cumulative energy used by HPC facilities up to time t
$E_{site}(t)$	Cumulative energy used by HPC facilities on-site up to time t
$E_{source}(t)$	Cumulative energy used by HPC facilities from the source of electricity up to time t
ϵ	Transmission efficiency of energy from source to HPC. Can be obtained from utility companies/energy regulatory commissions. Typically ranges from 90% to 95% efficiency
η_{site}	Amount of water withdrawn on-site per kWh of energy consumed by HPC. It is the sum of η_{evap} and $\eta_{blowdown}$. Can be obtained from ASHRAE Handbook or research papers. According to a 2016 study by Lawrence Berkeley National Laboratory, the value in the US is 1.8 L/kWh.
η_{evap}	Amount of water lost to evaporation per kWh of energy consumed on-site. Can be obtained from cooling tower manufacturers, ASHRAE, or the Cooling Technology Institute (CTI). EPA estimates it range from 1.2 L/kWh to 1.8 L/kWh.
$\eta_{blowdown}$	Amount of water lost due to blowdown per kWh of energy consumed on-site. Can be obtained from facility-specific water quality measurements allowing us to find value of CoC. Apply the formula: $\eta_{blowdown} = \frac{\eta_{evap}}{CoC-1}$
CoC	Cycles of Concentration; Number of times water is cycled through system before a limit that requires blowdown is reached. Can be obtained from cooling system manufacturers/operational logs from facilities. Usually ranges from 3 to 7.
η_{source}	Amount of water used by electricity utility per kWh of electricity generated. Can be obtained from energy mix data from utility providers/regional energy reports
γ	Annual growth rate of energy consumption. Can be obtained from historical data analysis/industry reports from the International Data Corporation (IDC). IDC estimates the annual growth rate to be around 19.5%
ρ	The fraction of water that is recycled and reused on-site (Recycling efficiency). This value varies with different facilities but you could obtain the value by checking facilities' water reuse strategies
$W_{blowdown}(t)$	Cumulative volume of blowdown water from cooling systems up to time t
δ	Concentration of pollutants in the blowdown water in mg/L. This value can be obtained from chemical usage testings like the material safety data sheets (MSDS). This varies depending on water treatment chemicals and local water quality
ϕ	Efficiency of the wastewater treatment process (fraction between 0 and 1). Can be obtained from operational performance metrics from treatment plants or Discharge Monitoring Reports (DMRs). Ranges from 75% to 99%
$M_{poll}(t)$	The cumulative mass of pollutants discharged into the environment up to time t

5.3 Assumptions

1. **We assume that energy consumption of HPC facilities grows at an exponentially over time at a constant annual growth rate.**

Justification: Exponential growth is a common pattern observed in technological advancements and data processing demands. This is supported by many forecasts and will allow us to represent the cumulative energy consumed more easily.

2. **We assume that the rate of water consumption is directly proportional to the rate of energy consumption and cooling efficiency.**

Justification: HPC facilities require significant cooling to maintain optimal operating temperatures. The energy used for computing generates heat, which need to be cooled with cooling systems that consume water. As energy consumption increases, water consumption for cooling also increases proportionally.

3. **The water efficiency parameters for on-site operations, such as the amount of water lost to evaporation (η_{evap}) and the amount lost due to blowdown ($\eta_{blowdown}$) remain constant over time. The recycling efficiency (ρ) of water as well as the wastewater treatment efficient (ϕ) also are constant over time**

Justification: Unless there are significant technological upgrades or changes in operational practices and recycling technologies or treatment facilities, these efficiencies are unlikely to change significantly.

5.3.1 Water Consumption Model

We will first model the water consumption. The rate of water consumption is proportional to the rate of energy consumption and cooling efficiency. On-site and source, we have

$$\begin{aligned}\frac{dW_{site}}{dt} &= \eta_{site} \cdot \frac{dE_{site}}{dt} \\ \frac{dW_{source}}{dt} &= \eta_{source} \cdot \frac{dE_{source}}{dt}\end{aligned}$$

$E_{source}(t)$ is equal to $\frac{E_{site}(t)}{\epsilon}$ because of transmission and distribution losses. Thus, $\frac{dW_{source}}{dt} = \eta_{source} \cdot \frac{1}{\epsilon} \cdot \frac{dE_{site}}{dt}$. From differentiating $W_{total}(t) = W_{site}(t) + W_{source}(t)$, we get $\frac{dW_{total}}{dt} = \frac{dW_{site}}{dt} + \frac{dW_{source}}{dt}$. We can now substitute our $\frac{dW_{site}}{dt}$ and $\frac{dW_{source}}{dt}$ to get

$$\frac{dW_{total}}{dt} = \eta_{site} \cdot \frac{dE_{site}}{dt} + \eta_{source} \cdot \frac{1}{\epsilon} \cdot \frac{dE_{site}}{dt} = \left(\eta_{site} + \frac{\eta_{source}}{\epsilon} \right) \left(\frac{dE_{site}}{dt} \right).$$

We can now integrate both sides of the equation from 0 to t with respect to time to get

$$\begin{aligned}\int_0^t \left(\frac{dW_{total}}{dt} \right) dt &= W_{total}(t) \text{ and} \\ \int_0^t \left[\left(\eta_{site} + \frac{\eta_{source}}{\epsilon} \right) \left(\frac{dE_{site}}{dt} \right) \right] dt &= \left(\eta_{site} + \frac{\eta_{source}}{\epsilon} \right) (E_{site}(t) - E_{site}(0)).\end{aligned}$$

Let E_0 be the current on-site energy consumption of HPCs which we estimated to be around 324 TWh. We assume that energy consumption for HPCs will grow exponentially. Thus, we can say that $E_{site}(t) = E_0 e^{\gamma t}$. We can then subtract E_0 from both sides to get $E_{site} - E_0 = E_0(e^{\gamma t} - 1)$. Now substitute back in the equation we integrated to get

$$W_{total}(t) = \left(\eta_{site} + \frac{\eta_{source}}{\epsilon} \right) (E_0(e^{\gamma t} - 1))$$

However, this total doesn't account for reusing the gray water. To account for this, we will update our η_{site} to be reduced by the fraction of water that is recycled. We can define a new variable $\eta_{site,eff} = \eta_{evap} + (1 - \rho)(\eta_{blowdown})$ in order to account for recycling. By definition, $\eta_{blowdown} = \frac{\eta_{evap}}{C_{oc}-1} - \eta_{drift \text{ loss}}$. However, because drift loss is negligible, we just have $\eta_{blowdown} = \frac{\eta_{evap}}{C_{oc}-1}$. Thus, we have $\eta_{site,eff} = \eta_{evap} + (1 - \rho) \left(\frac{\eta_{evap}}{C_{oc}-1} \right) = \eta_{evap} \left(\frac{C_{oc}-\rho}{C_{oc}-1} \right)$.

Also, using an average for η_{source} without considering the energy mixes can overestimate or underestimate the actual source-based water consumption. Energy could come from coal, natural gas, nuclear, wind, and others. Each of these has its own water consumption intensities. For example, coal uses 72.62 liters/kWh while natural gas uses 10.61 liters/kWh. In order to calculate η_{source} , we need to sum the products of the energy mix and water consumption intensities:

$$\eta_{source} = \sum_i (\text{Energy Mix Percentage}_i)(\text{Water Consumption Intensity}_i).$$

Let $\frac{\eta_{source}}{\epsilon}$ be equal to the effective amount, $\eta_{source,eff}$. So, now we have that the water amount consumed by HPCs as

$$W_{Total}(t) = (\eta_{site,eff} + \eta_{source,eff})(E_0(e^{\gamma t} - 1))$$

With this, we can assess the significance of HPC water use relative to global water resources.

5.3.2 Pollution Model

The amount of water that is blowdown from on-site and not recycled (water that is drained from cooling equipment), $W_{blowdown}(t)$, is equal to $(1 - \rho) \cdot \eta_{blowdown}(E_0 e^{\gamma t})$. This water is treated before it is released into the environment. Let $M_{poll,total}(t)$ be the total mass of pollutants before the water is treated. $W_{poll,total}(t) = \delta \cdot W_{blowdown}(t)$. From this, we can compute the mass of pollutants after treatment as $M_{poll}(t) = (1 - \phi) \cdot M_{poll,total}(t)$. Substituting, we have:

$$M_{poll}(t) = (1 - \phi) \cdot \delta \cdot (1 - \rho) \cdot \eta_{blowdown}(E_0 e^{\gamma t}).$$

With this total pollution mass discharged, we can evaluate HPC's contribution to global environmental degradation.

5.4 Environmental Impact of Water Usage

Water cooling makes extreme computing possible. Over 90% of new HPC data centers are water-cooled today.^[21] These facilities consume vast amounts of water, which could strain local water resources, especially in regions where there is high water scarcity. The discharge of pollutants from cooling system blowdown can also cause environmental pollution, adversely affecting water quality and harming aquatic ecosystems. Thus, it is important to develop effective cooling for HPC facilities. Understanding water usage can help optimize cooling systems, and monitoring pollutant discharge is important for meeting environmental regulations.

Water consumption is directly proportional to energy consumption. Any increase in computational demand leads to higher energy use. Thus, this results in higher water usage. This aspect also affects areas including infrastructure investments and sustainability goals. Choices will have to be made on energy sources, and facilities will have to align energy consumption strategies with water conservation efforts to support sustainability.

6 Results and Actions

6.1 Technical Solutions and Policy-Oriented Solutions

In order to reduce carbon emissions associated with the energy usage of HPC on a worldwide scale, infrastructure must be put into place to accommodate renewable energy usage. Some ways to do this would be to increase energy efficiency in data centers through liquid cooling, or optimizing code to run faster. Facilities can implement algorithms that schedule HPC workloads based on the availability of renewable energy. For example, facilities can run non-critical computations during periods of high renewable energy. Facilities can also consider implementing heat reuse systems that capture and reuse waste heat generated by equipment to heat nearby buildings which would offset some of their carbon footprint. In order to reduce the water consumption and pollution impacts of HPC, facilities can invest in implementing gray water systems to reuse water from sinks and other non-industrial sources for cooling after appropriate treatment. Similarly, facilities can invest in advanced water treatment systems to increase the recycling rate of blowdown water which leads to less polluted water that is discharged. HPC facilities can also move towards using renewable resources. They can either source their electricity from renewable energy providers or install their own on-site renewable energy generators, both of which will allow for higher sustainability.

Governments can set clear corporate sustainability regulations in order to reduce water consumption and pollutant over specific timeframes. They could also encourage or require all facilities to get the ISO 14001 certification which implements an environmental management system to manage and reduce impacts. Governments can also incentivize sustainable practices by providing grants, tax credits, or rebates for investments in sustainable technology like technical solutions mentioned earlier. Instead of incentivization, governments could also penalize facilities that consume large amounts of non-renewable energy with taxes.

6.2 Investments in Advanced Water Treatment Systems

By increasing the investments in water treatment systems, our ρ and ϕ from model 4b will increase. Let's first consider the water consumption model. As our ρ value increases, more water that is blowdown is recycled and so $\eta_{site,eff}$ will decrease. As $\eta_{site,eff}$ decreases, the total water consumption, $W_{total}(t)$ decreases proportionally. Thus total water consumption decreases over time because of improved efficiency. This means that less water is withdrawn from external sources.

For the pollution model, the mass of pollutants is given by $M_{poll}(t) = (1 - \phi) \cdot \delta \cdot (1 - \rho) \cdot \eta_{blowdown}(E_0 e^{\gamma t})$. As ρ and ϕ increase, $M_{poll}(t)$ gets smaller. Thus, the mass of pollutants discharged decreases, reducing the environmental impact of the facility. You can further see the effect of advancing treatment systems by computing the change in water consumption and change in pollution mass at time t .

7 Strengths and Weaknesses

7.1 Strengths

A major strength of our climate model is that it models the carbon concentration over time in the land biosphere, ocean, and atmosphere, which all contribute to different effects worldwide. Using scientific principles to model the transfer of carbon between each of these sectors, we have made sure that our model can accurately represent reality to some degree. Our model is a simplified version of the Bern Simple Climate Model, which has been tested on collected data and cross-checked with other climate models to ensure accuracy. Using this model as our baseline helped us greatly in the creation of the model, as we were able to reference BernSCM if we had an issue with our alteration. We also utilized key concepts like

impulse response functions to try to account for the delayed effect of carbon moving through the biosphere, which allowed our model to more accurately represent reality. Our model is also easy to extend, since additional factors that affect carbon content change in the atmosphere, ocean, land biosphere or temperature could just be added to the respective equation.

The water consumption and pollution models effectively quantify the environmental impacts of HPC facilities by establishing a direct link between energy consumption and water usage, including both on-site and source-based factors. This model is strong because of its comprehensive scope and adaptability to different operational parameters. By incorporating variables like recycling efficiency and treatment process, we can easily see the sustainability effects when we improve in certain areas. The model is also backed by extensive research and uses methodologies supported by several environmental agencies. By providing reliable predictions, the model aids in informed decision-making and advancement of sustainable practices within the HPC industry.

7.2 Weaknesses

Our climate model makes many assumptions that, if taken into account, would greatly enhance the model and its accuracy. Specifically, we assumed the atmosphere, ocean surface, and land biosphere were generally homogenous in terms of carbon content, this is not true as there are higher carbon concentrations around cities and areas that release more carbon into the atmosphere as well as larger carbon deposits in rainforests. Using one number to represent the carbon content throughout the world could greatly undermine the problem in certain areas, making our results less useful as they only represent averages. In addition, we assume that deep sea heat transfer and carbon intake is negligible on the time scale we are considering, which is true, but this makes applying our model beyond this time frame unreasonable without making those additions. Our results would also benefit from additional exchanges between each sector, like a land biosphere to ocean factor or an ocean biosphere factor, but these violate our model's assumption of linearity in the system (which is why we were able to use impulse response factors).

We also encountered issues with implementation. At first, we wanted to use the concept of fugacity to replace partial pressure of CO_2 in the ocean since the BernSCM uses a fifth degree polynomial in m_S to calculate it. However, our fugacity equation was only dependent on temperature, and since the magnitude of temperature was relatively low compared to ocean content the rate of change of the carbon content of the ocean was unbalanced and grew without bound. So, we decided to use the BernSCM implementation of change in dissolved carbon content in the ocean, which was much more reliable and tested to fit existing data. Our code also exposed the complexity of our model when numerically determining carbon content. To accurately handle the impulse response functions, we needed to integrate from negative infinity to t at every time step. This made computation very slow so we changed the lower bound to a large negative number. This didn't seem to change the results too much, but it would have improved our model if we tried to analytically solve the system of differential equations and used the closed forms instead of numerical approximation. This would also allow us to see trends directly instead of observing a linear or exponential pattern.

The water consumption and pollution models have limitations because of its reliance on simplifying assumptions and static parameters like constant exponential growth rates and operational efficiencies which may not hold true over time due to technological advancements. The model does not account for environmental interactions and instead only provides water consumption and pollution level. The accuracy of the model also greatly depends on accurate input data for constants like the efficiencies. Obtaining facility-specific data can be difficult due to proprietary restrictions, leading to potential errors or over-reliance on industry averages that may not reflect local conditions. While the model is backed by research, it may not fully capture regional regulatory differences, societal factors, or economic considerations.

8 Conclusion

There is no doubt that humanity must act to reduce the environmental impact of technological advancements. This statement is entirely in self-interest: our negligence will be our demise if we do not take action, when we are gone the ecosystems of the Earth will return to normal. In this paper we created mathematical models to describe the environmental impacts of one of the hottest new technological advancements of today: High-Powered Computing. Our model reported major effects through temperature increase, carbon content in the atmosphere, ocean, and land biosphere, as well as ocean acidification and sea level increase up to 2030. The growth of HPC and the resulting increased demand for energy is a cause for concern as we have not developed the infrastructure to support renewable energy on a worldwide scale. We hope that this paper has provided valuable insight into the environmental impact of HPC growth, and that it has inspired the action that is required to prevent potential catastrophic consequences.

9 Works Cited

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10 Appendix A: Matlab Code

```

1 % MATLAB script to solve and plot system of 4 ODEs
2 global mAO NPP0 mL0 mS0 em;
3 mAO = 3310;
4 NPP0 = 62.1; % PgC/yr = Gtc/yr, average over 2000-2018
5 mL0 = 470;
6 mS0 = 145.77; %2019, in PgC
7 em = 0.1085;
8
9 function dmdt = odesystem(t, m)
10     global em mAO;
11     dmdt = zeros(4, 1);
12     %dmdt(1) = em*(1.15)^t - dmdt(2) - dmdt(4); % atmosphere
13     dmdt(1) = 37.55 - em + em*(1.15)^t - dmdt(2) - dmdt(4);
14     dmdt(2) = (7.51e14)*m(1) - (1.007e14)*new(m); % ocean
15     dmdt(3) = (3.67e-8)*log(m(1)/mAO) - (8.44e-9)*m(3); % temp
16     dmdt(4) = NPP(m) + impulse(t,m); %land
17     disp(m);
18 end
19 function f = fugacity(m)
20     f = exp(5.511 + (5.2723e-2)*(18.3+m(3)) - (2.731e-4)*(18.3+m(3))^2);
21 end
22 function n = new(m)
23     n=DIC(m)*exp(.0423*m(3));
24 end
25 function d = DIC(m)
26     b = 4.58*m(2);
27     d = 1.3*b + (3.77e-3)*b^2 + (0.924e-5)*b^3 + (0.136e-7)*b^4 + (1.258e-10)*b^5;
28 end
29 function r = delay(t) % derivative of r(t) wrt time
30     r = 0.72*exp(-t/2.18) - 0.703*exp(-t/2.86) - .0134*exp(-t/20) - .00293*exp(-t/100);
31 end
32 function I = impulse(t, m)
33     t1 = linspace(-10000, t, 10000);
34     delay_values = delay(t - t1);
35     NPP_values = NPP(m);
36     I = trapz(t1, NPP_values .* delay_values);
37 end
38
39 function N = NPP(m)
40     global NPP0 mAO;
41     N = NPP0*(1 + 0.2*log(m(1)/mAO));
42 end
43
44 tspan = [0 6];
45
46 % Initial conditions
47 m0 = [mAO; mS0; 0; mL0];
48
49 % Solve the system using ode15
50 [t, m] = ode15s(@odesystem, tspan, m0);
51
52 subplot(1,4,1);
53 plot(t, m(:, 1), 'r', 'LineWidth', 1.5);
54 xlabel('Time (t), years', 'FontSize', 12);
55 ylabel('Atmosphere', 'FontSize', 12);
56 title('Atmosphere Carbon', 'FontSize', 14);
57 grid on;
58
59 % Plot Ocean
60 subplot(1,4,2);
61 plot(t, m(:, 2), 'g', 'LineWidth', 1.5);
62 xlabel('Time (t), years', 'FontSize', 12);
63 ylabel('Ocean', 'FontSize', 12);
64 title('Ocean Carbon', 'FontSize', 14);
65 grid on;
66
67 % Plot Temperature
68 subplot(1,4,3);
69 plot(t, m(:, 3), 'b', 'LineWidth', 1.5);
70 xlabel('Time (t), years', 'FontSize', 12);

```

```
71 ylabel('Temperature Change (C)', 'FontSize', 12);
72 title('Temperature', 'FontSize', 14);
73 grid on;
74
75 % Plot Land
76 subplot(1,4,4);
77 plot(t, m(:, 4), 'k', 'LineWidth', 1.5);
78 xlabel('Time (t), years', 'FontSize', 12);
79 ylabel('Land Biosphere', 'FontSize', 12);
80 title('Land Carbon', 'FontSize', 14);
81 grid on;
```